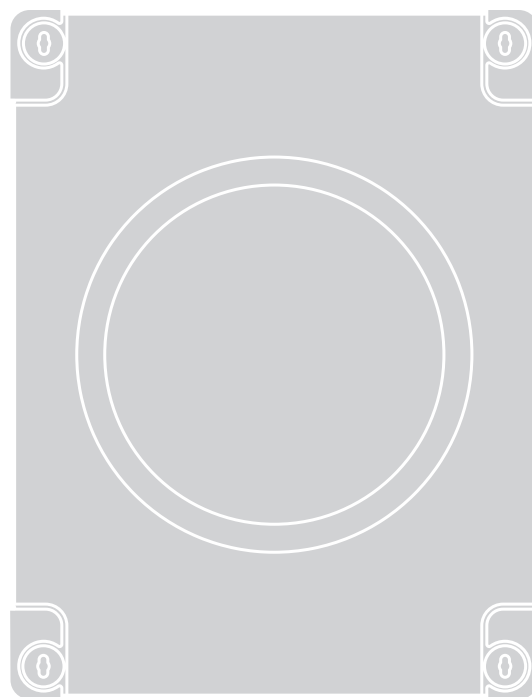


Nice

MC424L



Control unit



IT - Istruzioni ed avvertenze per l'installazione e l'uso

EN - Instructions and warnings for installation and use

The Nice logo is located in the bottom right corner, consisting of the word "Nice" in white text on a black square background.

Nice

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ing, impact, dropping or contact with any type of liquid. Never place the product near sources of heat or expose to naked flames. This may damage product components and cause malfunctions, fire or hazardous situations. If this occurs, suspend installation immediately and contact the Nice Service Centre.

- Never make modifications to any part of the product. Operations other than as specified can only cause malfunctions. The manufacturer declines all liability for damage caused by makeshift modifications to the product.
- The product should not be used by children or people with impaired physical, sensorial or mental capacities or who have not received adequate training in the safe use of the product.
- Do not allow children to play with the control devices for the automation system. Keep remote controls out of reach of children
- Children must be supervised to make sure that they do not play with the device.
- On the power line to the system, install a device for disconnection from the power mains with a gap between contacts that assures complete disconnection in the conditions of overvoltage category III.
- Connect the control unit to an electric power line equipped with an earthing system.
- The product's packaging materials must be disposed of in full compliance with local regulations.

1 PRODUCT DESCRIPTION

The MC424L control unit has been designed to control Wingo, Too, S-fab 24 V electromechanical actuators, for automated swing gates or doors.

IMPORTANT! – Any uses other than those specified herein or in environmental conditions other than as stated in this manual are to be considered improper and are strictly prohibited!

The MC424L control unit operates on the basis of a current sensitivity system which checks the load of the motors connected to it. The system automatically detects travel stops, memorises the running time of each motor and recognises obstacles during normal movement. This feature makes installation easier as there is no need to adjust the working times nor the leaf delay.

The control unit is pre-programmed for the functions usually required and incorporates a radio receiver for the remote controls. Using a simple procedure it is also possible to choose more specialised functions (see chapter 5).

The control unit is provided with a type SM socket for plug-in radio receivers (see paragraph "6.1 - Connection of a radio receiver") and a type IBT4N socket which, through the IBT4N interface, allows BusT4 devices to be connected as well as the Oview programming unit (see paragraph "6.2 - Connection of IBT4N interface").

The control unit is set up to be powered by PS124 back-up batteries as an emergency power supply, in the event of a power cut (see paragraph "6.3 - Connection of back-up battery mod. PS124"); it is also set up to be connected to the "Solemyo" solar energy power supply system (see paragraph "6.4 - Connection of Solemyo system").

2 INSTALLATION

In order to explain certain terms and aspects of an automatic 2-leaf swing door or gate system refer to the typical system shown in **fig 1**.

Key to fig. 1:

1. Wingo, Too, S-fab 24 V electromechanical actuator
2. Wingo, Too, S-fab 24 V electromechanical actuator
3. Lucy24 flashing light
4. Key-operated selector switch
5. "PHOTO" pair of photocells
6. "FOTO1" pair of photocells
7. "PHOTO2" pair of photocells
8. Control unit.

In particular, please note that:

- Refer to the product instructions for the characteristics and connection of the photocells.
- Activation of the "PHOTO" pair of photocells have no effect on the gate during opening, while they reverse movement during closing.
- Activation of the "PHOTO 1" pair of photocells stops both the opening and closing manoeuvres.
- Activation of the "PHOTO2" pair of photocells (connected to the suitably programmed AUX input) has no effect during closing while they invert movement during opening.

GENERAL SAFETY WARNINGS AND PRECAUTIONS

Safety warnings

- **IMPORTANT! – This manual contains important instructions and warnings for personal safety.** Incorrect installation could cause serious physical injury. Read all parts of the manual carefully before starting work. If in doubt, interrupt installation and contact the Nice Service Centre for clarifications.
- **IMPORTANT! – Important instructions: keep this manual in a safe place to enable future product maintenance and disposal procedures.**

Installation warnings

- Before commencing installation, check that the product is suitable for the intended kind of use (see paragraph 2.2 "Limits of use" and chapter "Product technical specifications"). If not suitable, do NOT proceed with installation.
- During installation, handle the product with care, avoiding the risk of crush-

To check the parts of the control unit see **fig. 2**.

Key to fig. 2:

- A 24 V AC power supply socket
- B Back-up PS124 battery / Solemyo solar energy power supply system socket (for further details see chapter 6.3)
- C Services fuse (2A, type F)
- D "SM" connection for radio receiver
- E Motor terminal M1 (part first closed)
- F Motor terminal M2 (part first opened)
- G Flashing lights output terminal
- H OGI or electric locking output terminal
- I 24 V DC terminals for services and phototest
- L9..L13 Input LEDs
- OK "LED OK" LED status
- L1..L8 Programming LEDs
- LR Radio programming LED
- M Terminals for inputs
- N Terminal for radio aerial
- O Motor selector switch
- Q Connector for IBT4N
- R Network fuse (see: technical specifications of product)
- S Network supply (L-Line; N-Neutral) (see: technical specifications of product)
- T Connection to earth
- U Pressacavo
- P1..P3 Buttons to programme control unit
- P4 Button to programme radio

2.1 - Preliminary checks for installation

Before proceeding with installation, check the condition of the product components, suitability of the selected model and conditions of the intended installation environment:

- Ensure that all conditions of use remain within the limits of product application and within the "Product technical specifications".
- Ensure that the selected installation environment is compatible with the overall dimensions of the product (**fig. 3**).
- Ensure that the selected surfaces for product installation are solid and guarantee a stable fixture.
- Make sure that the fixing zone is not subject to flooding. If necessary, mount the product raised from the ground.
- Ensure that the space around the product enables easy and safe completion of manual manoeuvres.
- Make sure that the automation is provided with mechanical stops on both closing and opening.

2.2 - Product application limits

The product may be used exclusively with WG2024, WG3524, WG4024, WG5024, TOO3024, TOO4524, XME2024 gearmotors.

2.3 - Installation

To install the control unit, proceed as shown in **fig. 4**. Also observe the following warnings:

- The control unit is supplied in an enclosure that if correctly installed assures an IP54 protection rating. The control unit is therefore suitable for installation outdoors.
- Fix the control unit to a flat, vertical, non-removable surface that is adequately protected from potential impacts. **Important!** – The bottom of the control unit must be at least 40 cm from the ground.
- Insert the dedicated cable clamps or pipe glands into the lower part of the enclosure (**fig. 4**). **Important!** – If the cable protection tubes end in a pit, it is likely that condensation will form inside the control unit, which will damage the electronic board. In this case, protect the control unit adequately so as to prevent the formation of condensation.
- The cable clamps can be inserted on the long side of the enclosure only if the control unit is installed in a protected indoor environment.

To install the other devices present in the automation, refer to the relevant instruction manuals.

2.4 - Electrical connections

IMPORTANT!

- All electrical connections must be made with the unit disconnected from the mains power supply and with the buffer battery disconnected, if present in the automation.
- Connections must be made exclusively by qualified personnel.
- Make sure that all the electric cables used are of a suitable type.

- 01. Loosen the screws of the cover.
- 02. Prepare the electrical cable routing holes.
- 03. Connect the cables, referring to the electrical diagram shown in **fig. 5**. To connect the electrical power supply cable, vedere **fig. 2** e **fig. 6** nei punti R, S, T, U. To secure the cable, use the **clip U**.
Note – To facilitate cable connections, the terminals can be removed from their seats.
- With the exception of the photocell inputs when the PHOTOTEST function is activated, if the inputs of the NC (Normally Closed) contacts are not in use they should be jumped with the "COMMON" terminal. Refer to paragraph 2.4.3 for further information.
- If there is more than one NC contact on the same input, they must be connected in SERIES.
- If the inputs of the NO (Normally Open) contacts are not used they should be left free.
- If there is more than one NO contact on the same input, they must be connected in PARALLEL.
- The contacts must be electromechanical and potential-free. Stage connections, such as those defined as "PNP", "NPN", "Open Collector", etc. are not allowed.
- in the case of overlapping doors, you can select which motor must start opening first by connecting it to terminal M2.

Key to figs. 2 - 5a - 5b - 5c:

Terminals	Function	Description	Type of cable
L - N -	Power supply line	Mains power supply	3 x 1,5 mm ²
M	Motor 1	M1 motor connection (Note 1)	3 x 1,5 mm ²
M	Motor 2	M2 motor connection	3 x 1,5 mm ²
1÷2	Flashing light	Connection of flashing light 24 V~ max 25 W	2 x 1 mm ²
3÷4	Open Gate indicator / Elect.Lock	Connection for Open Gate Indicator 24 V~ max 5 W or Electric lock12 V~ max 25 VA ("See chapter 5 - Programing")	OGI: 2 x 0,5 mm ² Electric lock: 2 x 1 mm ²
5	Common 24 V~ (with Everything in stand by / phototest)	Power Supply +24 V~ for TX photocells with phototest (max. 100 mA); "COMMON" for all inputs, safety, with "Everything in stand by" function activated (Note 2)	1 x 0,5 mm ²
6	0 V~	Power supply 0V~ for services	1 x 0,5 mm ²
7	24 V~	Power input for services, without "Everything in stand by" (24 V~ max 200 mA)	1 x 0,5 mm ²
8	Common 24 V~	Common for all inputs (+24 V~) without "Everything in stand by"	1 x 0,5 mm ²
9	ALT	Input with ALT function (emergency, safety shutdown) (Note 3)	1 x 0,5 mm ²
10	PHOTO	NC Input for safety devices (photocells, sensitive edges)	1 x 0,5 mm ²
11	PHOTO 1	NC Input for safety devices (photocells, sensitive edges)	1 x 0,5 mm ²
12	STEP BY STEP	Input for cyclical functioning (OPEN-STOP-CLOSE-STOP)	1 x 0,5 mm ²
13	AUX	Auxiliary input (Note 4)	1 x 0,5 mm ²
	Aerial	Connection for the radio receiver aerial	screened cable type RG58

- Note 1** – This is not used for single leaf gates (the control unit automatically recognises if only one motor has been installed).
- Note 2** – The "Everything in stand by" function serves to reduce consumptions. For further details on the electrical connections refer to paragraph 2.4.1 "Everything in stand by/Phototest connection" and for programming refer to chapter 5.2 "Programmable functions".
- Note 3** – The STOP input can be used for "NC" or constant resistance 8.2 kΩ contacts during auto-recognition (please refer to the "Programming" chapter).
- Note 4** – The AUX factory auxiliary input is programmed with the "Partial open type 1" function but can be programmed with any of the following functions:

Function	Input type	Description
PARTIAL OPEN TYPE 1	NO	Fully opens the upper leaf
PARTIAL OPEN TYPE 2	NO	Opens the two leaf half way

Function	Input type	Description
OPEN	NO	Only carries out the opening manoeuvre
CLOSE	NO	Only carries out the closing manoeuvre
PHOTO 2	NC	PHOTO 2 function
STOP	NA	Stops the manoeuvre
DISABLED	—	No function

2.4.1 - Notes about connections

Most connections are extremely simple and many of them are direct connections to a single user point or contact. The following figures show examples of how to connect external devices:

• Everything in stand by / Phototest connection

The “Everything in stand by” function is active as standard. It is excluded automatically only when the Phototest function is activated. **Note** - The “Everything in stand by” and Phototest functions are alternatives as one excludes the other.

The “Everything in stand by” function allows consumptions to be reduced. Three types of connections can be obtained:

- with “Everything in stand by” active (**energy saving**); see electrical diagram in **fig. 5a**
- standard connection: without “Everything in stand by” and without “Phototest”; see electrical diagram in **fig. 5b**
- without “Everything in stand by” and with “Phototest”; see electrical diagram in **fig. 5c**

When the “Everything in stand by” function is active, 1 minute after the end of a manoeuvre the control unit goes into “Everything in stand by”, turning off the Inputs and Outputs to reduce consumptions. The status is indicated by the “OK” LED which begins to flash more slowly. **WARNING** – If the control unit is powered from a photovoltaic panel (“Solemyo” system) or a buffer battery, the “Everything in stand by” function must be activated as shown in the electrical diagram in **fig. 5a**. When the “Everything in stand by” function is not required, the “Phototest” function may be activated. This verifies at the beginning of a manoeuvre that the connected photocells operate correctly. To use this function, first connect the photocells appropriately (see electrical diagram in **fig. 5c**) and then activate the function. **Note** – When the phototest is activated, the inputs subjected to the test procedure are PHOTO, PHOTO1 and PHOTO2. If one of these inputs is not used it must be connected to terminal no. 8.

• Key switch connection

Example 1 (fig. 7a): How to connect the switch in order to perform the STEP-BY-STEP and STOP functions

Example 2 (fig.7b): How to connect the switch in order to perform the STEP-BY-STEP and one of the auxiliary input functions (PARTIAL OPENING, OPEN ONLY, CLOSE ONLY ...)

Note – For electrical connections with the “Everything in stand by” function active, see “Everything in stand by/Phototest function” in this paragraph 2.4.1.

• Connection for Gate Open Indicator / Electric lock (fig. 8)

If the gate open indicator has been programmed, the output can be used as an open gate indicator light. The light, flashes slowly during opening and quickly during closing; If it is on but does not flash, this indicates that the gate is open. If the light is off, the gate is closed. Se the output has been programmed as an electric lock, it is activated for 3 seconds each time opening begins.

2.4.2 - STOP type input

The MC424L control unit can be programmed for two types of STOP input:

- **NC type STOP** for connecting up to NC type contacts.
- **Constant resistance STOP**. It enables the user to connect up to the control unit of devices with 8.2k Ω constant resistance (e.g. sensitive edges). The input measures the value of the resistance and disables the manoeuvre when the resistance is outside the nominal value. Devices with normally open “NO” or normally closed “NC” contacts, or multiple devices, even of different types, can be connected to the constant resistance STOP input, provided that appropriate adjustments are made; see Table 1.

WARNING! – If the constant resistance STOP input is used to connect devices with safety functions, only the devices with 8.2 K Ω constant will resistance output guarantee the fail-safe category 3.

TABLE 1				
second device type:	1st device type:			
	NO	NC	8,2 K Ω	
	In parallel (note 1)	(note 2)	In parallel	
	NC (note 2)	In series (note 3)	In series	
8,2 K Ω	In parallel	In series	(note 4)	

Notes to Table 1:

Note 1 – Any number of NO devices can be connected to each other in parallel, with an 8.2 K Ω termination resistance (**fig. 9a**). For electrical connections with the “Everything in stand by” function active, see “Everything in stand by/Phototest function” in this paragraph 2.4.1.

Note 2 – The NO and NC combination can be obtained by placing the two contacts in parallel, and placing an 8.2 K Ω resistance in series with the NC contact. It is, therefore, possible to combine 3 devices: NO, NC and 8.2 K Ω (**fig. 9b**).

Note 3 – Any number of NC devices can be connected in series to each other and to an 8.2 K Ω resistance (**fig. 9c**).

Note 4 – Only one device with an 8.2 K Ω constant resistance output can be connected; multiple devices must be connected “in cascade” with a single 8.2 K Ω termination resistance (**fig. 9d**).

2.5 - Initial start-up and electrical connections

IMPORTANT! – Connections must be made exclusively by qualified personnel.

After powering up the control unit, check that all the LEDs flash rapidly for a few seconds, then perform the following checks:

1. Check that there is a voltage of approximately 30Vdc on terminals 6-7. If not, unplug the unit immediately and carefully check the connections and input voltage.
2. After initially flashing rapidly, the Led OK will indicate the control unit is working correctly by flashing regularly at 1 second intervals. When there is a variation in the inputs, the Led OK led will rapidly flash twice to show that the input has been recognised.
3. If the connections are correct, the LED for the “NC”-type inputs will be on, while those for the “NO” type inputs must be off. See **fig. A** and **Table 2a**.

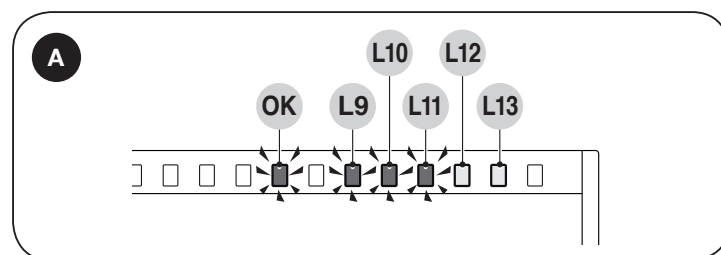


TABLE 2a		
INPUT	INPUT TYPE	STATUS LED
STOP	STOP NC	L9 On (Solo dopo punto 5)
	CONSTANT RESISTANCE STOP 8.2 K Ω	L9 On (Solo dopo punto 5)
PHOTO	NC	L10 On
FOTO1	NC	L11 On
STEP-BY-STEP	NO	L12 Off
AUX	OPEN PARTIALLY type 1 - NO	L13 Off
	OPEN PARTIALLY type 2 - NO	L13 Off
	OPEN ONLY - NO	L13 Off
	CLOSE ONLY - NO	L13 Off
	FOTO2 - NC	L13 On

4. Check that the relative LEDs switch on and off when the devices connected to the inputs are operated.
5. Check that pressing keys P2-P3 for 3 seconds makes both motors do a short opening manoeuvre with the upper door's motor starting first. Block the manoeuvre by pressing a key. If the motors do not start an opening, reverse the polarity of the motor cables. If the first motor to move is not the upper door's, swap M1 with M2. LED L9 lights up to indicate that auto-recognition was successful.

2.6 - Motor selector switch

The control unit has a selector switch (see **fig. 2 - point O**) for specifying the type of motor used, as shown in **Table 2b**. No other configurations are valid.

TABLE 2b	
Type of motor	Motor selector switch
Generic	

TABLE 2b	
Type of motor	Motor selector switch
WG2024	
WG3524	
WG4024	
WG5024	
TOO3024	
TOO4524	
XME2024	

2.7 - Automatic search system for the limit switches

On the successful completion of the various controls, start the automatic search system phase for the limit switches. This work is necessary as the MC424L control unit must “measure” how long the opening and closing manoeuvres take. This procedure is completely automatic and detects the mechanical opening and closing stops by measuring the load on the motors.

are enabled (STOP, PHOTO and PHOTO1). The procedure will be immediately interrupted if a safety device triggers or a command arrives. **The leafs MUST be positioned at approximately mid-travel.**

Procedure – At the same time, press buttons P2 (Stop/Set) and P3 (Close ▼) (fig. 2) to start the search stage that includes:

- Check and save the motor selector switch.
- Check and save the ALT input (NC / 8.2 KΩ).
- Both motors open briefly.
- Motor closes the lower leaf until it reaches the mechanical closing stop.
- The upper leaf motor closes until it reaches the mechanical closing stop.
- The motor of the upper leaf begins to open.
- After the programmed delay, opening of the lower leaf begins. If the delay is insufficient, block the search by pressing P2 (Stop/Set) (fig. 2), then modify the time (see chapter 5).
- The control unit measures the movement required for the motors to reach the opening mechanical stops.
- Complete closing manoeuvre. The motors can start at different times, the aim is to prevent the leafs from shearing by maintaining a suitable delay.
- End of the procedure with memorisation of all measurements.

All these phases must take place one after the other without any interference from the operator. If the procedure does not continue correctly, it must be interrupted with the P2 (Stop/Set) button. Repeat the procedure, modifying some parameters if necessary, for example the current sensitivity cut-in thresholds (see chapter 5). This procedure can be repeated without needing to delete the memory.

3.1 - Testing

The testing sequence refers to the control unit programmed with the preset functions. See paragraph 5.1:

- Make sure that the activation of the STEP-BY-STEP input generates the following sequence of movements: “Open, Stop, Close, Stop”.
- Make sure that the activation of the AUX input (Type 1 partial opening function) manages the “Open, Stop, Close, Stop” sequence of the motor of the upper leaf only, while the motor of the lower leaf remains in the closed position.
- Perform an opening manoeuvre and check that:
 - the gate continues the opening manoeuvre when PHOTO is engaged
 - the opening manoeuvre stops when PHOTO1 is engaged and only continues when PHOTO1 is disengaged
 - The manoeuvre stops when PHOTO2 (if installed) is engaged and the closing manoeuvre starts
- Make sure that the motor switches off when the door reaches the mechanical stop.
- Perform an opening manoeuvre and check that:
 - The manoeuvre stops when PHOTO is engaged and the opening manoeuvre starts
 - The manoeuvre stops when PHOTO1 is engaged and the opening manoeuvre starts again when PHOTO1 is disengaged
 - the gate continues the closing manoeuvre when PHOTO 2 is engaged
- Check that the stop devices connected to the ALT input trigger the immediate stopping of any movement in progress and a slight reversal
- Check that the level of the obstacle detection system is suitable for the application:
 - During both the opening and the closing manoeuvres, prevent the leaf from moving by placing an obstacle and check that the manoeuvre inverts before exceeding the force set down by law
- Other checks may be required depending on which devices are connected to the inputs.

Warning – If an obstacle is detected as moving in the same direction for 2 consecutive manoeuvres in the same direction, the control unit partially inverts both motors for just 1 second. At the following command, the leafs begin the opening manoeuvre and the first current sensitivity cut-in for each motor is considered as a mechanical stop during the opening cycle. The same happens when the mains power supply is switched on: the first command is always an opening manoeuvre and the first obstacle is always considered as a mechanical stop during the opening cycle.

3.2 - Commissioning

Commissioning can only be performed after positive results of all test phases.

- 1 Prepare the automation technical documentation, which must contain the following documents: overall drawing of the automation, electrical wiring diagram, risk assessment and relative solutions adopted (refer to the relevant forms on our website www.niceforyou.com), manufacturer’s declaration of conformity for all devices used and installer’s declaration of conformity.
- 2 Affix a dataplate on the gate, specifying at least the following data: type of automation, name and address of manufacturer (responsible for commissioning), serial number, year of construction and CE mark.
- 3 Before commissioning the automation, ensure that the owner is adequately informed of all associated risks and hazards.

3 TESTING AND COMMISSIONING

These are the most important phases of automation set-up for ensuring maximum system safety. The test can also be performed as a periodic check of automation devices. Testing and commissioning of the automation must be performed by skilled and qualified personnel, who are responsible for the tests required to verify the solutions adopted according to the risks present, and for ensuring observance of all legal provisions, standards and regulations, and in particular all requirements of the standard EN 12445, which establishes the test methods for checking automations for doors and gates. The additional or optional devices must undergo a specific test for functionality and correct interaction with MC424L. Refer to the instruction manuals of the individual devices.

4 TROUBLESHOOTING (troubleshooting guide)

4.1 - Signals with flashing light

Some devices are set up to give signals by which it is possible to recognise the operating status or any eventual faults. If a flashing light is connected to the FLASH output on the control unit, it will flash every second while a manoeuvre is being carried out. If any faults are detected, the flashing light will flash more quickly; this will be repeated twice, with a 1 second pause between. **Table 3a** describes the cause and solution for each type of signal.

TABLE 3a - Signals from OK LED and flashing light		
Flashes	Problem	Solution
2 short red flashes 1 second pause 2 short red flashes	Intervention of a photocell	One or more photocells do not permit the movement or while it is in progress have reversed the movement; check if any obstacles are present.
3 short red flashes 1 second pause 3 short red flashes	Intervention of "Obstacles Detected" function by power limiter	During movement the motors have encountered greater effort; check the cause and if necessary increase the power level of the motors.

TABLE 3a - Signals from OK LED and flashing light

Flashes	Problem	Solution
4 short red flashes 1 second pause 4 short red flashes	Intervention of STOP input	At the start of the manoeuvre, or during movement, the devices connected to the STOP input have intervened; check the cause.
5 short red flashes 1 second pause 5 short red flashes	Error in control unit's internal parameters	Wait at least 30 seconds and then try again to give a command and if necessary also turn off the power supply; if the status remains there could be a serious failure and the electronic circuit board may need to be replaced.
6 short red flashes 1 second pause 6 short red flashes	The maximum number of consecutive manoeuvres or manoeuvres per hour has been exceeded	Wait a few minutes for the manoeuvre limiter to go back under the maximum limit.
7 short red flashes 1 second pause 7 short red flashes	Fault on the electrical circuits	Wait at least 30 seconds and then try again to give a command and if necessary also turn off the power supply; if the status remains there could be a serious failure and the electronic circuit board may need to be replaced.
8 short red flashes 1 second pause 8 short red flashes	There is already a command present that means further commands cannot be carried out	Check the nature of the command still to be carried out; for example it might be a command from a clock on the OPEN input.

4.2 - Signals on the control unit

There is a set of LEDs on the control unit (see **fig. 2**.) each of which can give specific signals, both in normal operation and in the event of a fault. **Tables 3b** and **3c** describe the cause and solution for each type of signal.

TABLE 3b - LEDs on control unit's terminals

All LEDs	Problem	Solution
No LED is on	No power to the control unit	Check whether there is power to the control unit: measure a voltage of approx. 30 V DC (or 24 V DC if powered by battery) on terminals 6-7. Check the 2 fuses, if the OK LED isn't on or flashing either, there is probably a serious failure so the control unit will have to be replaced.
OK LED	Problem	Solution
Off	Fault	Check whether there is power; check that the fuses have cut in; if necessary, check the cause of the fault and then replace them with others of the same value
On	Serious fault	There is a serious fault; try turning off the control unit for a few seconds; if the status continues, there is a fault and the electronic circuit board must be replaced
1 flash per second	All OK	Normal operation of the control unit
1 flash every 5 seconds	All OK	Control unit on stand by
2 fast flashes	There has been a variation in the status of the inputs	This is normal when there is a change in one of the inputs: OPEN, STOP, intervention of the photocells or use of the remote control
Series of flashes separated by a one-second pause	Various	It's the same signal as on the flashing light (see TABLE 3a - Signals of OK LED)
ALT LED	Problem	Solution
Off	Intervention of the ALT input	Check the devices connected to the ALT input
On	All OK	ALT input active
PHOTO LED	Problem	Solution
Off	Intervention of PHOTO input	Check the devices connected to the PHOTO input
On	All OK	PHOTO input active
PHOTO1 LED	Problem	Solution
Off	Intervention of PHOTO1 input	Check the devices connected to the PHOTO1 input
On	All OK	PHOTO1 input active
Sbs LED	Problem	Solution
Off	All OK	Sbs input not active
On	Intervention of Sbs input	This is normal if the device connected to the Sbs input is actually active
AUX LED	Problem	Solution
Off	All OK	AUX input not active
On	Intervention of AUX input	This is normal if the device connected to the AUX input is actually active

TABLE 3c - LEDS on control unit's keys

LED L1	Description
Off	During normal operation this indicates "Automatic closure" is not active
On	During normal operation this indicates "Automatic closure" is active
LED L2	Description
Off	During normal operation this indicates "Close again after photo" is not active
On	During normal operation this indicates "Close again after photo" is active
Flashing	Function programming in progress
LED L3	Problem
Off	During normal operation this indicates "Always Close" is not active
On	During normal operation this indicates "Always Close" is active
Flashing	Function programming in progress. If it flashes at the same time as LED L4 the position recognition stage must be carried out (see paragraph " 2.6 - Automatic search for stops")

TABLE 3c - LEDS on control unit's keys

LED L4	Problem
Off	During normal operation this indicates "Stand-By" is active
On	During normal operation this indicates "Phototest" is active
Flashing	Function programming in progress. If it flashes at the same time as LED L3 the position recognition stage must be carried out (see paragraph " 2.6 - Automatic search for stops")
LED L5	Problem
Off	During normal operation this indicates output OGI as OGI (open gate indicator)
On	During normal operation this indicates output OGI as ELS (electric locking system)
Flashing	Function programming in progress
LED L6	Problem
Off	During normal operation this indicates "Pre-flashing" is not active
On	During normal operation this indicates "Pre-flashing" is active
Flashing	Function programming in progress
LED L7	Problem
Off	During normal operation this indicates "Building" is not active
On	During normal operation this indicates "Building" is active
Flashing	Function programming in progress
LED L8	Problem
Off	During normal operation this indicates "Light gates" is active
On	During normal operation this indicates "Heavy gates" is active
Flashing	Function programming in progress

4.3 - Maintenance notice

The control unit lets the user be notified when to carry out a maintenance check on the automation system. The number of manoeuvres after which the signal is given can be selected from 8 levels, via the adjustable parameter, "Maintenance notice" (see **Table 6**).

The adjustments are set based on the number of manoeuvres carried out.

The maintenance request signal is made via Flash.

Based on the number of manoeuvres carried out compared to the programmed limit, Flash and the maintenance control light give the signals shown in **Table 3d**.

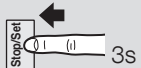
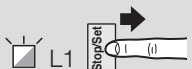
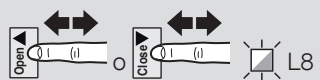
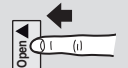
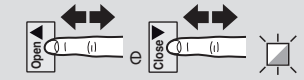
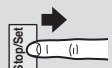
TABLE 3d - Maintenance notice signals

Number of manoeuvres	Signals on Flash	Signal on maintenance control light
Under 80% of the limit	Normal (0.5 seconds on, 0.5 seconds off)	On for 2 seconds when opening begins
Between 81 and 100% of the limit	At the start of the manoeuvre it stays on for 2 seconds then proceeds as usual	Flashes for the whole manoeuvre
Over 100% of the limit	At the start and end of the manoeuvre it stays on for 2 seconds then proceeds as usual	Constant flashing

4.4 - Historic faults log

The control unit can display any eventual faults that have been detected in the last 8 manoeuvres, e.g. the interruption of a manoeuvre through the intervention of a photocell or a sensitive edge. To check the faults log, proceed as in **Table 3e**.

TABLE 3e - Historic faults log

01. Hold down key P2 (Stop/Set) for approx. 3s	
02. Release key P2 (Stop/Set) when LED L1 starts to flash	
03. Press and release keys P1 (Open ▲) or P3 (Close ▼) to move the flashing of the LED to L8 for the "Faults list" parameter	
04. Hold down key P2 (Stop/Set) , and keep it down throughout steps 5 and 6	
05. Wait for approx. 3 seconds, after which LED L1, corresponding to the outcome of the most recent manoeuvre, will come on	
06. Press and release keys P1 (Open ▲) and P3 (Close ▼) to select the desired manoeuvre: The corresponding LED will give a number of flashes equal to those normally emitted by the flashing light after a fault (see Table 3a)	
07. Release key P2 (Stop/Set)	

5 PROGRAMMING

The MC424L control unit has some programmable functions. These functions are pre-set in a typical configuration that should suffice for most automations. The functions can be changed at any time, either before or after the automatic stops search stage, through appropriate programming procedures described below.

5.1 - Preset functions

- Automatic closing: enabled
- Condominium function: disabled
- Pre-flashing: disabled
- Close after photo: disabled
- Opening delay: level 5 (20%)
- Everything in stand by / Phototest: Everything in stand by
- Electric Lock / Gate open indicator: Electric Lock
- STOP input: auto recognition NC type / 8,2K Ω
- Heavy gates: disabled
- Pause time: 20 seconds
- Auxiliary input: type 1 partial opening (only the upper leaf motor is activated)
- Current sensitivity: Level 3

5.2 - Programmable functions

To ensure the system is best suited to the user's requirements, and safe in the various different conditions of use, the MC424L control unit offers the possibility to programme several functions or parameters, as well as the function of a number of inputs and outputs.

5.2.1 - First level programming

First level functions (ON-OFF functions)

The programmable functions available are arranged over 2 levels.

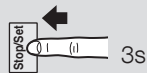
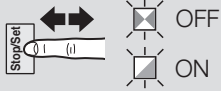
The first level functions can be adjusted in ON-OFF mode (active or not active); in this case each LED L1..L8 indicates a function, if it's on the function is active, if it's off the function is not active; see **Table 4**. For the programming procedure see **Table 5**.

TABLE 4 - First level functions (ON / OFF)

LED	Function	Description
L1	Automatic closing	<u>Function ON:</u> after an opening manoeuvre, there is a pause (equal to the "pause time" set at 30 seconds that can be changed to 5, 15, 30, 45, 60, 80, 120, 180 seconds) after which the control unit automatically starts up a closing manoeuvre. <u>Function OFF:</u> the operation is "semi-automatic".
L2	Close again after photo	<u>Function ON:</u> if the photocells intervene during the opening or closing manoeuvre, the pause time is reduced to 4 seconds irrespective of the programmed "pause time". With "automatic closing" deactivated, if the photocells intervene during the opening or closing manoeuvre, "automatic closing" is activated with the "pause time" (4 sec.).
L3	Always close	<u>Function ON:</u> should there be a power cut, even a short one, 10 seconds after power is restored, the control unit detects the open gate and automatically starts a closing manoeuvre, preceded by 5 seconds of pre-flashing.
L4	Stand-by / Phototest	<u>Function ON:</u> phototest. Instead of "All on Stand by", the "Phototest" function can be activated, which checks the correct operation of the connected photocells at the start of a manoeuvre. The photocells must be connected correctly for this function to be used (see electrical diagram in fig. 5c) and then the function activated. <u>Function OFF:</u> stand-by. The control unit has the "All on Stand by" function pre-set; if this is active, 1 minute after a manoeuvre ends, the control unit turns off the "All on Stand by" output (terminal 5), all the Inputs and the other outputs, so as to reduce consumption (see electrical diagram in fig. 5a). This function is compulsory if the control unit is powered exclusively by Solemyo photovoltaic panels. It is recommended even if the control unit is powered by mains electric and there is a desire to increase emergency operation with back-up battery PS124.
L5	Electric Locking / OGI (Open Gate Indicator)	<u>Function ON:</u> OGI (Open Gate Indicator). If the function is activated, terminals 3-4 can be used to connected an open gate indicator (24 V). <u>Function OFF:</u> electric locking. If the function is non activated, terminals 3-4 can be used to connect the electric locking (12 V).
L6	Pre-flashing	<u>Function ON:</u> the flashing light is activated 3 seconds before the start of the manoeuvre to give advance warning of a dangerous situation. <u>Function OFF:</u> The light starts to flash when the manoeuvre begins.
L7	Building function	<u>Function ON:</u> each command received triggers an opening manoeuvre that cannot be interrupted by other command pulses. <u>Function OFF:</u> each command received triggers OPEN-STOP-CLOSE-STOP. This behaviour is useful when a lot of people are using the automation with remote controls.
L8	Light/Heavy Gates	<u>Function ON:</u> if the function is activated, the control unit provides the option of managing heavy gates by using a different setting for the acceleration ramps and deceleration speeds during closure. <u>Function OFF:</u>

First level programming (ON-OFF functions)

IMPORTANT: The programming procedure sets a maximum time of 10 seconds between pressing one button and the next. After this time, the procedure ends automatically, saving the changes made up to that moment.

TABLE 5 - First level programming procedure	
01. Press and hold down key P2 (Stop/Set) for about 3 seconds	
02. Release the key when LED L1 starts to flash	
03. Press key P1 (Open ▲) or P3 (Close ▼) to move the flashing LED to the LED that represents the function to be changed	
04. Press key P2 (Stop/Set) to change the status of the function: (short flashes = OFF - long flashes = ON)	
05. Wait for 10 seconds (maximum time) to exit programming.	

Note: to programme other functions to "ON" or "OFF", while the procedure is being carried out, points 03 and 04 must be repeated during the same stage.

5.2.2 - Second level programming

Second level functions (adjustable parameters)

The second level parameters can be set to one of 8 values and are factory set as shown in grey in **Table 6**.

To change the parameter values, follow the steps in **Table 7**.

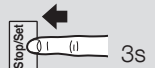
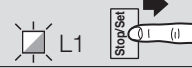
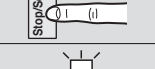
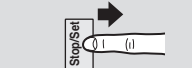
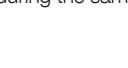
TABLE 6 - Second level functions (8 possible parameter values)				
Entry LED	Parameter	LED (level)	Value	Description
L1	Pause Time	L1	5s	Adjusts the pause time, i.e. the time before automatic re-closing. It is effective only if automatic closing is active
		L2	15s	
		L3	30s	
		L4	45s	
		L5	60s	
		L6	80s	
		L7	120s	
		L8	180s	
L2	AUX input	L1	Partial opening type 1	The control unit provides an auxiliary input that can be set up as one of the following 6 functions. Partial opening type 1: carries out the same function as the STEP-BY-STEP input, causing only the upper door to open. It only works with the gate completely closed, otherwise the command is interpreted as if it were a STEP-BY-STEP command. Partial opening type 2: carries out the same function as the STEP-BY-STEP input, causing the opening of the two doors for half of the time anticipated for total opening. It only works with the gate completely closed, otherwise the command is interpreted as if it were a STEP-BY-STEP command. Open Only: this input only carries out the opening with the Open-Stop-Open-Stop sequence. Close Only: this input only carries out the closing with the Close-Stop-Close-Stop sequence. Photo 2: carries out the function of the "PHOTO 2" safety device. Excluded: the input does not control any function.
		L2	Partial opening type 2	
		L3	Open only	
		L4	Close only	
		L5	Photo 2	
		L6	Stop (stops the manoeuvre)	
		L7	Excluded	
		L8	Excluded	
L3	Speed of motors	L1	Very slow	Adjusts the speed of the motors during normal running.
		L2	Slow	
		L3	Medium	
		L4	Fast	
		L5	Very fast	
		L6	Extremely fast	
		L7	Open "fast"; close "slow"	
		L8	Open "extremely fast", Close "medium"	
L4	Running down of motors after Closure	L1	No running down	Adjusts the duration of the "brief reversal" of both motors, after the closing manoeuvre has been carried out, with the aim of reducing the final residual thrust.
		L2	0.2s	
		L3	0.4s	
		L4	0.6s	
		L5	0.8s	
		L6	1.0s	
		L7	1.2s	
		L8	1.4s	

TABLE 6 - Second level functions (8 possible parameter values)

Entry LED	Parameter	LED (level)	Value	Description
L5	Strength of engines (amperometric sensitivity)	L1	Level 1 - Minimum strength	Adjusts the strength of both engines. The control unit has a system for measuring the current consumed by the two motors that is used to detect the mechanical stops and any eventual obstacles during movement of the gate. Since the current consumed depends on variable conditions (gate weight, various types of friction, gusts of wind, variations in voltage, etc.) there is the option to change the intervention threshold. 8 levels are provided: level 1 is the most sensitive (minimum strength), level 8 the least sensitive (maximum strength). Increasing the value of the degree of amperometric sensitivity increases the deceleration speed in the manoeuvre's closing stage. WARNING! – The suitably adjusted "amperometric" function (together with other essential devices) can be useful for complying with European standards EN 12453 and EN 12445, which require the use of procedures and devices to limit strengths and dangers in the movement of automatic doors and gates.
		L2	Level 2 - ...	
		L3	Level 3 - ...	
		L4	Level 4 - ...	
		L5	Level 5 - ...	
		L6	Level 6 - ...	
		L7	Level 7 - ...	
		L8	Level 8 - Maximum strength	
L6	Door delay	L1	0%	Adjusts the delay in the starting of the lower door's motor; can be programmed as a percentage of the working time.
		L2	5%	
		L3	10%	
		L4	15%	
		L5	20%	
		L6	30%	
		L7	40%	
		L8	50%	
L7	Maintenance notice	L1	500	Adjusts the number of manoeuvres after which to signal the automation's maintenance request (see paragraph "4.3 - Maintenance notice")
		L2	1000	
		L3	1500	
		L4	2500	
		L5	5000	
		L6	10000	
		L7	15000	
		L8	20000	
L8	Faults list	L1	Outcome 1 ^a manoeuvre (the most recent)	Allows the type of fault that has occurred in the last 8 manoeuvres to be checked (see paragraph "4.4 - Historic faults log")
		L2	Outcome 2 ^a manoeuvre	
		L3	Outcome 3 ^a manoeuvre	
		L4	Outcome 4 ^a manoeuvre	
		L5	Outcome 5 ^a manoeuvre	
		L6	Outcome 6 ^a manoeuvre	
		L7	Outcome 7 ^a manoeuvre	
		L8	Outcome 8 ^a manoeuvre	

Second level programming (adjustable parameters)

IMPORTANT: The programming procedure sets a maximum time of 10 seconds between pressing one button and the next. After this time, the procedure ends automatically, saving the changes made up to that moment.

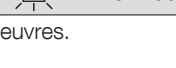
TABLE 7 - Second level programming procedure	
01. Press and hold down key P2 (Stop/Set) for about 3 seconds	
02. Release the key when LED L1 starts to flash	
03. Press key P1 (Open ▲) or P3 (Close ▼) to move the flashing LED to the LED that represents the "input LED" of the parameter to be changed	
04. Press and hold down key P2 (Stop/Set) until the end of point 06	
05. Wait for about 3 seconds, after which the LED that represents the current level of the parameter to be changed will come on	
06. Press key P1 (Open ▲) or P3 (Close ▼) to move the LED that represents the value of the parameter	
07. Release key P2 (Stop/Set)	
08. Wait for 10 seconds (maximum time) to exit programming.	

Note: to programme multiple parameters, while the procedure is being carried out, operations from point 03 to point 07 must be repeated during the same stage.

5.3 - Deleting the memory

All data stored in the control unit can be deleted and the unit restored to factory settings.

Follow the steps in **Table 8**.

TABLE 8 - Procedure for deleting the memory	
01. Press and hold the keys P1 (Open ▲) and P3 (Close ▼) at the same time	
02. Release the keys when the programming LEDs light up (after approx. 3 seconds)	
03. If the procedure was successful, the programming LEDs flash quickly for 3 seconds	
The following data are deleted: ALT configuration, position of stops, programming of first and second level, number of manoeuvres. The saved remote controls are not deleted (see paragraph 5.4, procedures D and E).	

5.4 - Storing remote controls on built-in radio

The control has a built-in radio receiver compatible with all remote controls that use NICE radio coding protocols **FLO, FLOR, O-CODE** and **SMILO**.

Two methods of storing remote control keys

Among the procedures available for storing remote controls, some allow them to be stored in "standard" mode (or Mode 1) and others in "customised" mode (or Mode 2).

STANDARD storing (also called "Mode 1")

Procedures of this type enable all the keys on the remote control to be stored at the same time, while they are being carried out. The system automatically matches a factory pre-set command to each key, as shown in the following diagram:

Command	Key
No. 1 - Step-by-Step	Will be matched to key 1
No. 2 - AUX	Will be matched to key 2
No. 3 - OPEN	Will be matched to key 3
No. 4 - CLOSE	Will be matched to key 4 (if the key is present on the remote control).

CUSTOMISED storing (also called "Mode 2")

Procedures of this type enable a single key from those present on the remote control to be stored, matching it to one of the following available commands: **Step-by-step, AUX, OPEN, CLOSE**.

The choice of key and of the command to be matched to it is made by the installer, based on what the automation needs.

Number of remote controls that can be stored

The control unit receiver has 100 memory locations. A location can store either a single remote control (i.e. all of its keys and commands) or a single key with the relative command.

PROCEDURES

WARNING! - In order to be able to carry out Procedures A, B, C, D the control unit's memory must be unlocked. If it is locked, carry out Procedure G to unlock it.

PROCEDURE A - Storing ALL the keys of a single remote control (STANDARD mode or Mode 1)

01. On the control unit: press and hold down key P4 (radio) until the "radio" LED comes on; then release the key.
02. (within 10 seconds) On the remote control that is to be stored: hold down any key and wait for the "radio" LED to give 3(*) long flashes (= store successful). Then, release the key.
03. Note - After the 3 long flashes another 10 seconds are available to store another remote control (if so desired), starting from step 02.

At the end of the Procedure the remote control keys will have been matched to the commands described in Mode 1.

PROCEDURE B - Storing a SINGLE KEY of a remote control (CUSTOMISED mode or Mode 2)

01. Choose the command that is to be matched to the key to be stored: for No. 1 - "Step-by-step" press key P4 (radio) once for No. 2 - "AUX" press key P4 (radio) twice for No. 3 - "OPEN" press key P4 (radio) three times for No. 4 - "CLOSE" press key P4 (radio) four times
02. On the control unit: press and release key P4 (radio) the same number of times as the desired control, as shown alongside the command chosen in step 01.
03. (within 10 seconds) On the remote control: hold down the key to be saved and wait for the "radio" LED to give 3(*) long flashes (= store successful). Then, release the key.

04. Note - After the 3 long flashes another 10 seconds are available to store another key (if so desired), starting from step 01.

PROCEDURE C - Storing a remote control via another remote control that has already been stored (remote storing)

This procedure allows a new remote control to be stored by using a second remote control, already stored in the same control unit. This allows the new remote control to be given the same settings as those of the remote already stored. Carrying out this procedure does not require direct action on key P4 (radio) on the control unit, but simply working within its reception area.

- 01.** On the remote control **to be stored**: press and hold down the key that is to be stored.
- 02.** On the control unit: after a few seconds (about 5) the **"radio" LED** comes on. Then, release the remote control key.
- 03.** On the remote control that is **already stored**: press and slowly release the stored key that is to be copied; do this **3 times**.
- 04.** On the remote control **to be stored**: press and hold the same key you pressed for point 01 and wait for the "radio" LED to flash slowly **3(*)** times (= store successful); then release the key.

(*) - Notes to Procedures A, B and C: – the "radio" LED can also give the following signals:

- **1 fast flash**, if the remote control is already stored.
- **6 flashes**, if the remote control's radio code is not compatible with that of the control unit receiver.
- **8 flashes**, if the memory is full.

PROCEDURE D - Deleting a single remote control (if stored in Mode 1) or a single key of a remote control (if stored in Mode 2)

- 01.** On the control unit: press and hold down key **P4 (radio)**. After about 4 seconds the **"radio" LED** comes on steadily (continue to hold down the key).
- 02.** On the remote control to be deleted from the memory: press and hold down a key **(*)** until the **"radio" LED** (on the control unit) gives 5 fast flashes (or 1 flash if the remote control or the key is not stored).

(*) - Note: – If the remote control is stored in Mode 1 press any key and the control unit deletes the whole remote control. If the remote control is stored in Mode 2, the stored key that is to be deleted must be pressed. To delete more keys stored in Mode 2, repeat the whole process for each key that is to be deleted.

PROCEDURE E - Deleting ALL stored remote controls

- 01.** On the control unit: press and hold down key **P4 (radio)**. After about 4 seconds the **"radio" LED** comes on steadily (continue to hold down the key). After about 4 seconds the **"radio" LED** goes off (continue to hold down the key).
- 02.** When the **"radio" LED** starts to flash, count 2 flashes and get ready to release the key **exactly during the 3rd flash that follows**.
- 03.** During deletion the **"radio" LED** flashes quickly.
- 04.** Finally, the **"radio" LED** gives 5 long flashes to show the deletion has been successful.

PROCEDURE G - Locking (or Unlocking) the Memory

WARNING! - This procedure locks the memory, preventing Procedures A, B, C, D from being carried out.

- 01.** Disconnect the control unit from the power supply.
- 02.** On the control unit: press and hold down key **P4 (radio)**; then reconnect the power supply to the control unit (still holding down the key).
- 03.** After 5 seconds the **"radio" LED** gives 2 short flashes; then release the key.
- 04.** (within 5 seconds) On the control unit: repeatedly press key **P4 (radio)** to choose one of the following options:
LED off = Deactivation of the Lock on storing
LED on = Activation of the Lock on storing.
- 05.** After 5 seconds from the last press on the key, the **"radio" LED** gives 2 slow flashes to signal the end of the procedure.

6 FURTHER DETAILS: accessories

6.1 - Connection of type SM radio receiver

The control unit has a connector for fitting a 4-control radio card complete with SM slot. This remote control device functions by means of transmitters which act on the inputs as per the following table:

Output Receiver	Control unit input
N° 1	Step by step
N° 2	AUX (reset value: Partially Open 1)
N° 3	"Open only"
N° 4	"Close only"

For further details refer to the receiver's manual.

Warning: before connecting the radio receiver turn off the control unit, remove the plastic cover and check no saliva is present.

6.2 - Connection of IBT4N interface

The control unit is provided with an "IBT4N" socket for the IBT4N interface, which allows all devices provided with a BusT4 interface to be connected, such as, for example, the Oview programming unit and the Wi-Fi IT4WIFI interface.

Warning: before connecting the interface turn off the control unit, remove the plastic cover and make sure no saliva is present. Then connect it, pushing it right in, making sure it goes in easily and is upright.

6.2 - Connecting model PS124 buffer battery

PS124 buffer batteries can be used to supply the control unit in case of network blackouts. To install and connect the battery, proceed as shown in **fig. 10**.

6.3 - Connecting the Solemyo system

The control unit is designed to be powered with the "Solemyo" photovoltaic system (photovoltaic panel and 24 V battery). To connect the Solemyo battery to the control unit, use the socket on the control unit that is normally used for the buffer battery (see paragraph 6.2).

IMPORTANT!

- When the automation is powered by the "Solemyo" system, it **MUST NOT BE POWERED at the same time from the electrical mains**.
- The Solemyo system can be used only if the "Everything in stand by" function on the control unit is ON and the connections are as shown in the diagram in **fig. 5a**.

7 MAINTENANCE DU PRODUIT

La logique de commande MC424L, en tant que partie électronique, n'a besoin d'aucune maintenance particulière. Vérifier toutefois périodiquement, au moins tous les 6 mois, le bon fonctionnement de toute l'installation suivant ce qui est indiqué dans le chapitre 3.

MISE AU REBUT DU PRODUIT

Ce produit est partie intégrante de l'automatisme et doit donc être mis au rebut avec ce dernier.

Comme pour l'installation, à la fin de la durée de vie de ce produit, les opérations de démantèlement doivent être effectuées par du personnel qualifié. Ce produit est constitué de différents types de matériaux : certains peuvent être recyclés, d'autres doivent être mis au rebut. Informez-vous sur les systèmes de recyclage ou de mise au rebut prévus par les règlements, en vigueur dans votre pays, pour cette catégorie de produit.

Attention ! – certains composants du produit peuvent contenir des substances polluantes ou dangereuses qui pourraient avoir des effets nuisibles sur l'environnement et sur la santé des personnes s'ils étaient jetés dans la nature.

Comme l'indique le symbole ci-contre, il est interdit de jeter ce produit avec les ordures ménagères. Par conséquent, utiliser la méthode de la « collecte sélective » pour la mise au rebut des composants conformément aux prescriptions des normes en vigueur dans le pays d'utilisation ou restituer le produit au vendeur lors de l'achat d'un nouveau produit équivalent.



Attention ! – les règlements locaux en vigueur peuvent appliquer de lourdes sanctions en cas d'élimination illicite de ce produit.

TECHNICAL CHARACTERISTICS OF THE PRODUCT

WARNINGS: • All technical characteristics stated refer to an ambient temperature of 20°C (±5°C). • Nice S.p.a reserves the right to modify the product at any time, while maintaining the same functionalities and intended use.

Mains power supply	MC424L Control units: 230 V~ ±10% 50 ÷ 60 Hz; fuse: 1A type T MC424L/V1 Control units: 120 V~ ±10% 50 ÷ 60 Hz; fuse: 2A type T
Max absorbed power	170 W
Emergency power supply	for PS124 buffer batteries and for Solemyo solar kit
Maximum motor current:	3A (with a "level 6" current sensitivity cut in)
Service power output	24 V = 200 mA maximum current (the voltage can vary from 16 to 33 V =)
Phototest Output	24 V = 100 mA maximum current (the voltage can vary from 16 to 33 V =)
Flashing lamp output	for flashing lamp 24 V =, maximum power 25 W (the voltage can vary from 16 to 33 V =)
Gate open indicator output	for indicator lamps at 24 V = maximum power 5 W (the voltage can vary from 16 to 33V =) or electric locks 12 V~ 25 W
STOP Input	for NC contacts or constant resistance 8,2 KΩ +/- 25%
Working time	automatic detection
Pause time	programmable at 5, 15, 30, 45, 60, 80, 120, 180 seconds
Discharge time	programmable to 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4 seconds
Leaf delay in open cycle	programmable at 0, 5, 10, 15, 20, 30, 40, 50% of working time
Leaf delay in close cycle	automatic detection
1st motor output	for Wingo (WG2024, WG3524, WG4024, WG5024), Too (TOO3024, TOO4524), S-fab (XME2024)
2nd motor output	for Wingo (WG2024, WG3524, WG4024, WG5024), Too (TOO3024, TOO4524), S-fab (XME2024)
Max. cable length	power supply 30 m Solemyo solar power kit 3 m motors 10 m other inputs/outputs 30 m flashing light 10 m OGI 30 m electric lock 10 m aerial 20 m (recommended less than 3 m)
Radio receiver	"SM" type coupling for receivers SMXI, SMXIS, OXI (Mode I and Mode II)
Temperatura di esercizio	from - 20 to 55 °C
Protection rating	IP 54 with enclosure intact
Dimensions (mm)	310 x 232 x H 122
Weight (kg)	4,1

EU Declaration of Conformity

and declaration of incorporation of "partly completed machinery"

Note - The content of this declaration corresponds to that specified in the official document deposited at the Nice S.p.A. headquarters and, in particular, to the latest revised edition available prior to the publishing of this manual. The text herein has been re-edited for editorial purposes. A copy of the original declaration can be requested from Nice S.p.A. (TV) I.

Number: 296/MC424

Revision: 5

Language: EN

Manufacturer's Name: Nice s.p.a.
Address: Via Pezza Alta 13, Z.I. Rustignè, 31046 Oderzo (TV) Italy
Person authorised to compile the technical documentation: Nice s.p.a.
Type of product: Comand central a 2 motor 24V dc
Model / Type: MC424, MC424L
Accessories: Radio receiver OXI

The undersigned Roberto Griffa, in the role of Chief Executive Officer, declares under his sole responsibility that the product described above complies with the provisions laid down in the following directives:

- Directive 2014/30/EU (EMC), in accordance with the following harmonised standards: EN 61000-6-2:2005, EN 61000-6-3:2007+A1:2011

The product also complies with the following directives according to the requirements envisaged for "partly completed machinery" (Annex II, part 1, section B): Directive 2006/42/EC of the EUROPEAN PARLIAMENT AND COUNCIL of 17 May 2006 related to machinery and amending the Directive 95/16/EC (recast).

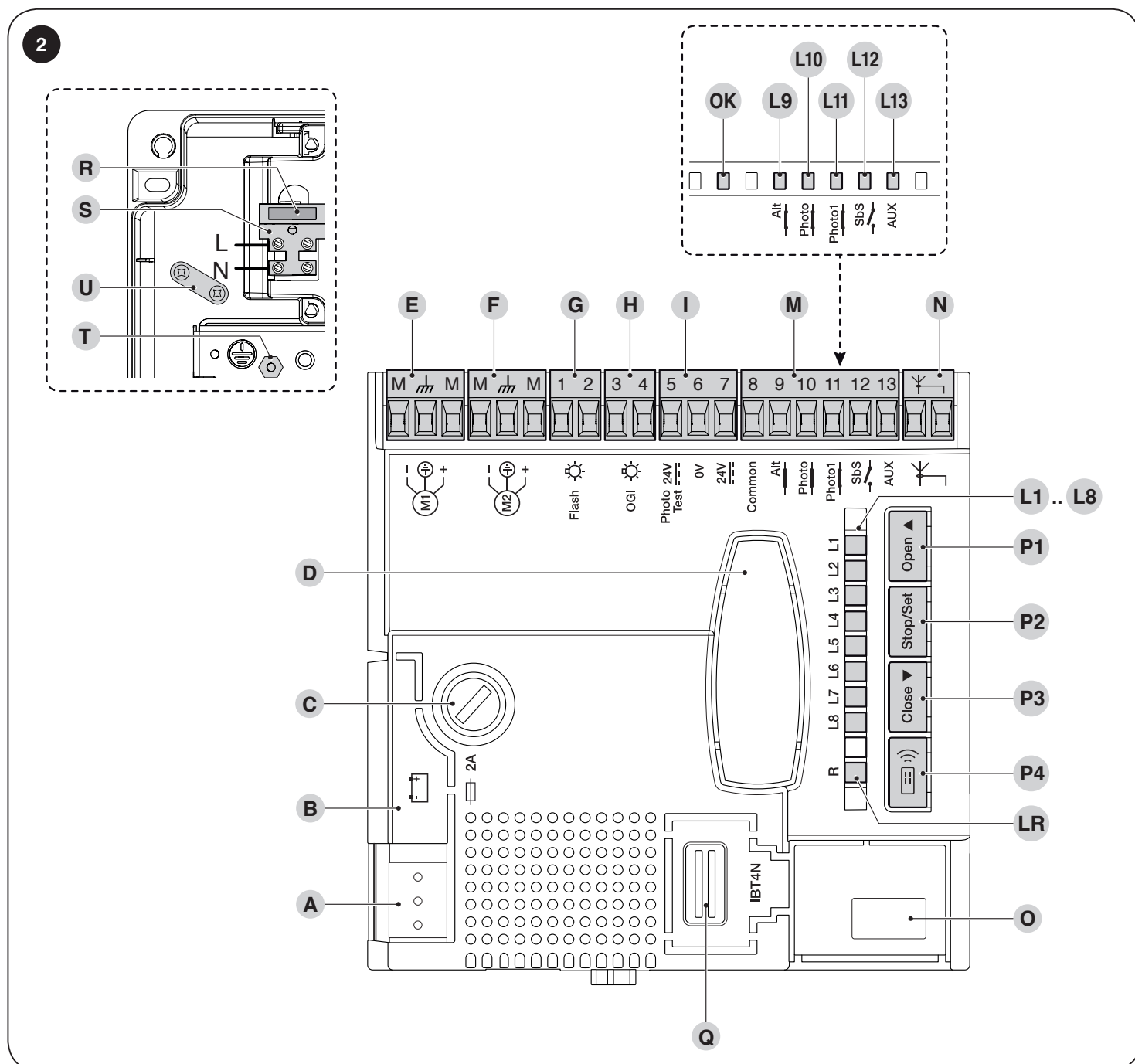
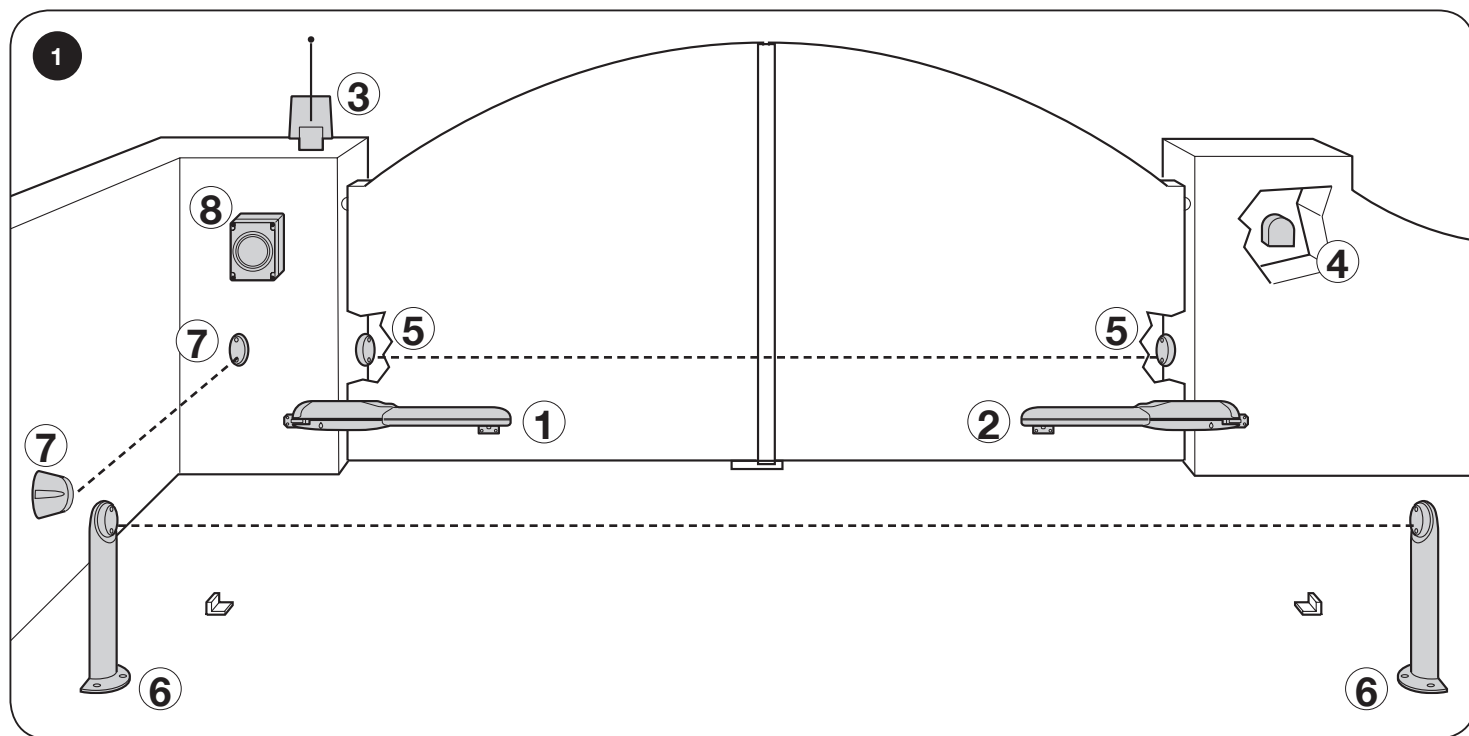
- It is hereby stated that the relevant technical documentation has been compiled in accordance with annex VII B of Directive 2006/42/EC and that the following essential requirements have been fulfilled: 1.1.1 - 1.1.2 - 1.1.3 - 1.2.1 - 1.2.6 - 1.5.1 - 1.5.2 - 1.5.5 - 1.5.6 - 1.5.7 - 1.5.8 - 1.5.10 - 1.5.1.
- The manufacturer undertakes to transmit to the national authorities, in response to a reasoned request, the relevant information on the "partly completed machinery", while maintaining full rights to the related intellectual property.
- Should the "partly completed machinery" be put into service in a European country with an official language other than that used in this declaration, the importer is obliged to arrange for the relative translation to accompany this declaration.
- The "partly completed machinery" must not be used until the final machine in which it is incorporated is in turn declared as compliant, if applicable, with the provisions of directive 2006/42/EC.

The product also complies with the following standards:
 EN 60335-1:2012+A11:2014, EN 62233:2008, EN 60335-2-103:2015

Oderzo, 05/09/2017

Ing. Roberto Griffa
 (Chief Executive Officer)

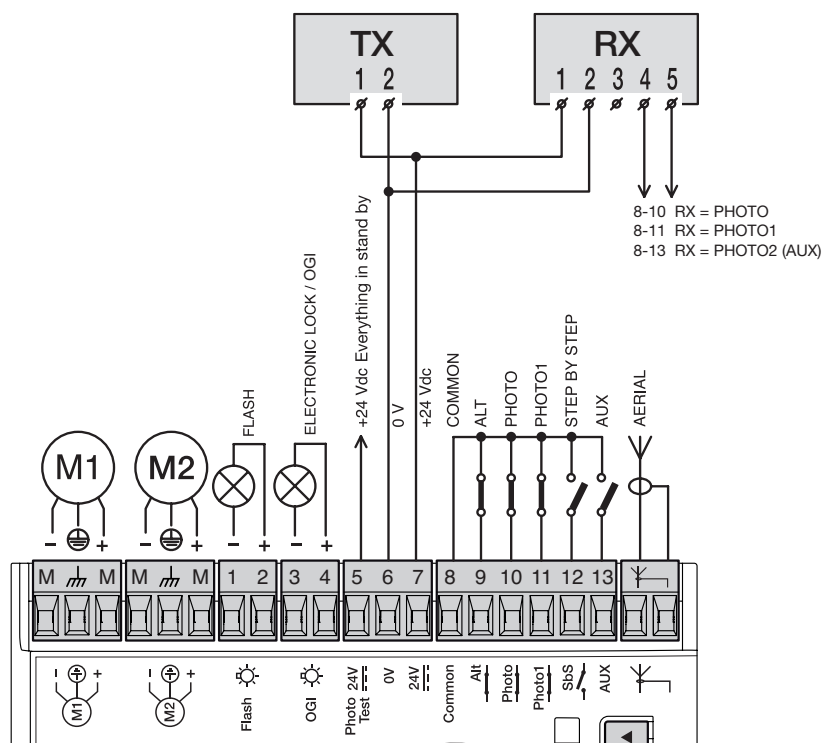




II – Italiano

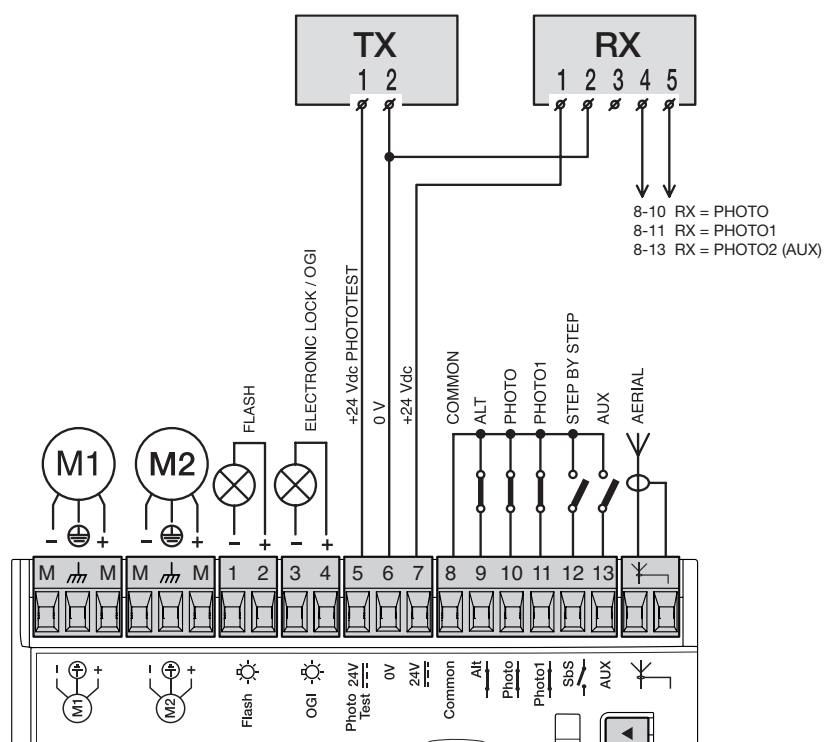
5b IT - Collegamento standard: senza utilizzare
"Stand by tutto" e senza "Fototest"

EN - Standard connection: without using
"Everything in stand by" or "Phototest"

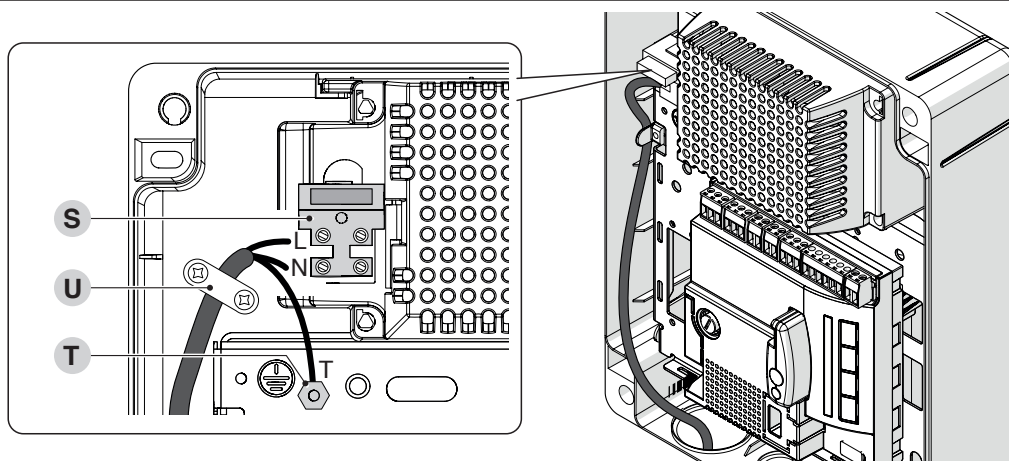


5c IT - Collegamento senza "Stand by tutto" con
"Fototest"

EN - Connection without "Everything in stand
by" with "Phototest"

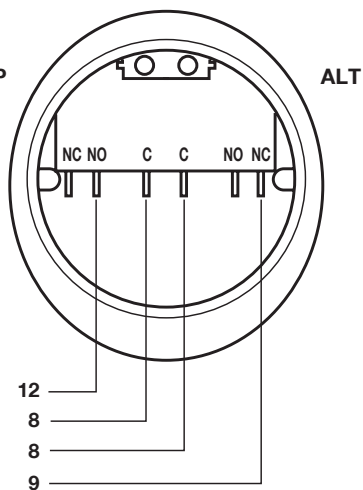


6



7a

STEP BY STEP

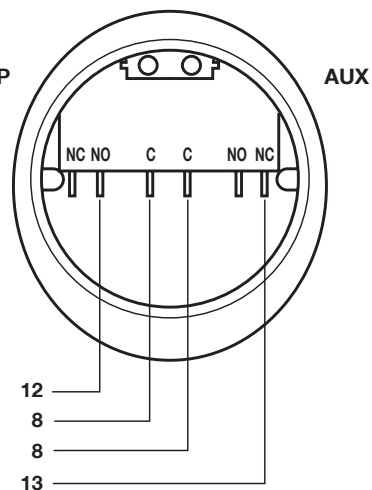


IT - Per il collegamento ALT, con “Stand by tutto” attiva, collegare il morsetto n° 5 e non il n° 8

EN - For the ALT connection with “Everything in stand by” active, connect terminal no. 5 and not no. 8

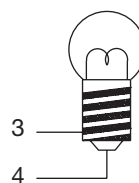
7b

STEP BY STEP

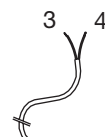


8

33 V=
max 5 W



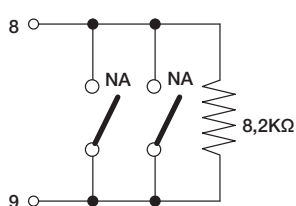
OGI



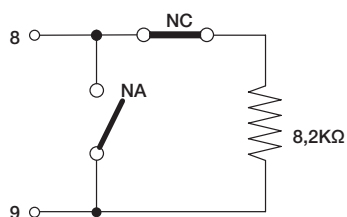
EL

12 V~
max 25 VA

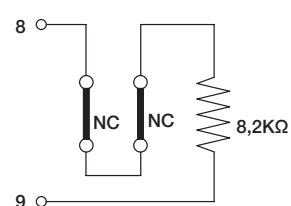
9a



9b



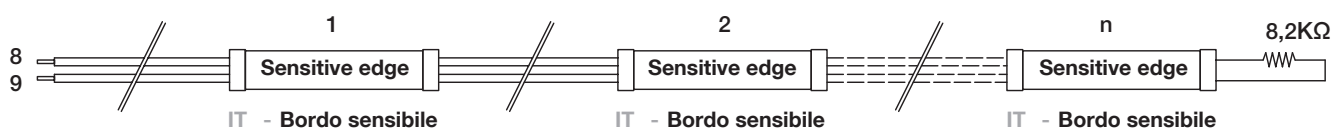
9c



IT - Con “Stand by tutto” attiva collegare il morsetto n° 5 e non il n° 8

EN - With “Everything in stand by” active connect terminal no. 5 and not no. 8

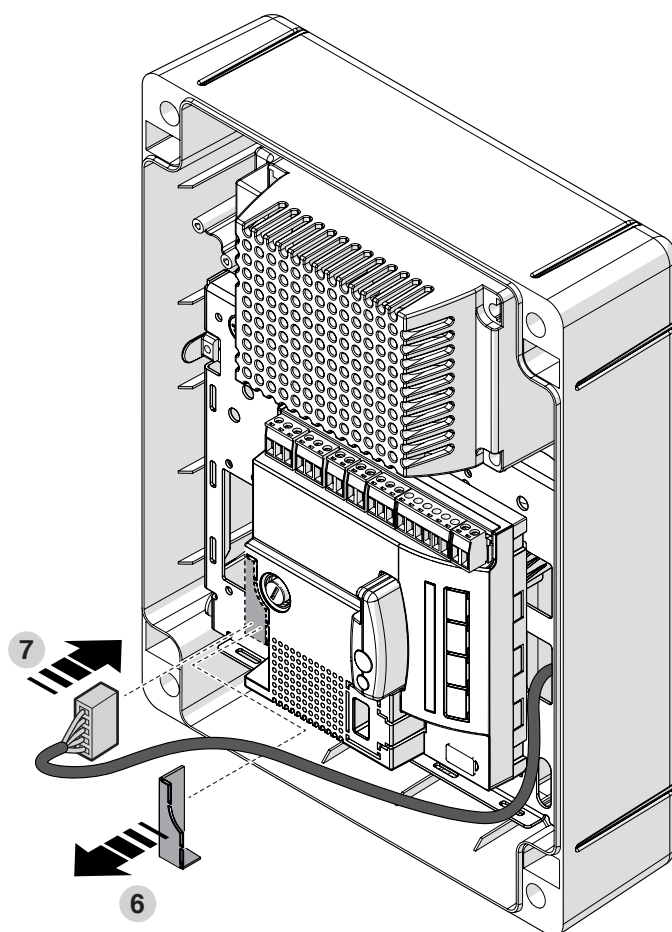
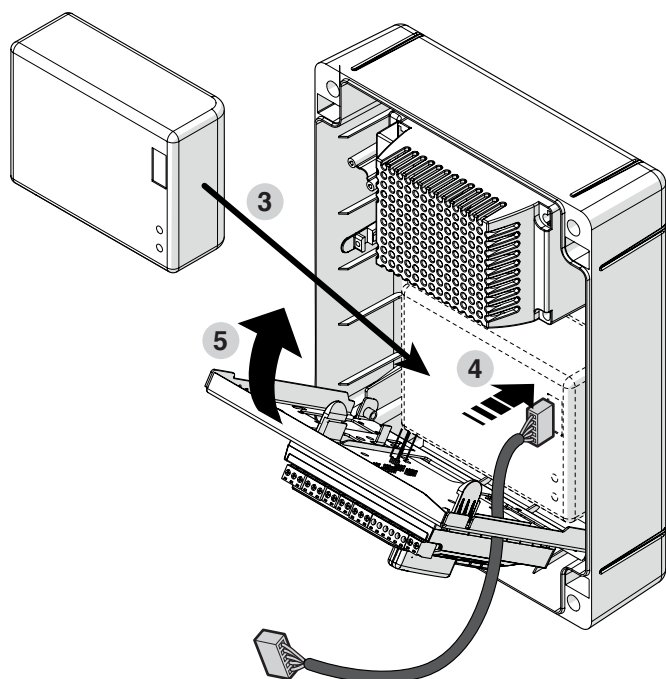
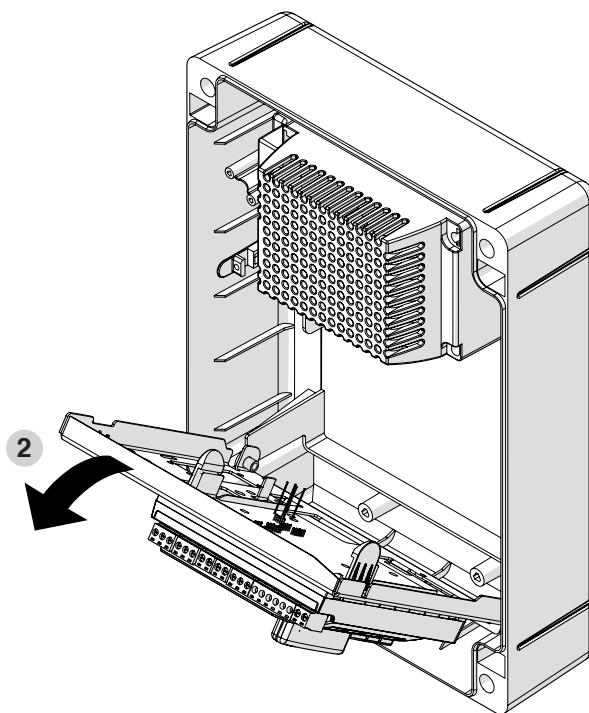
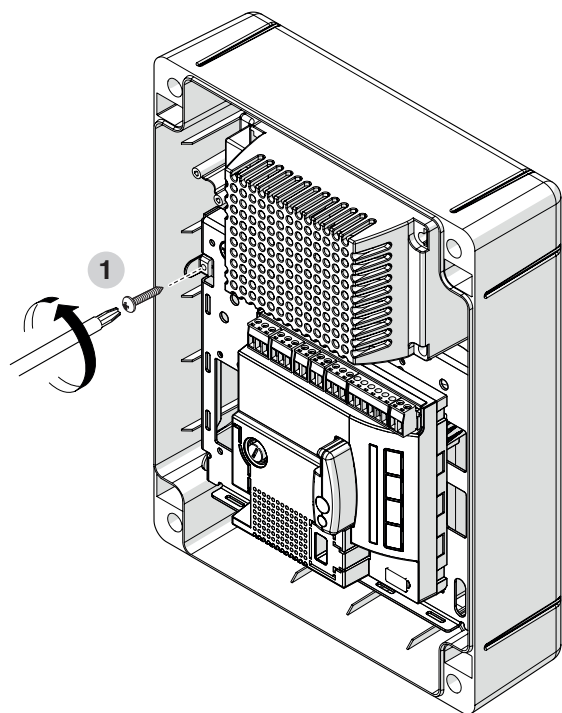
9d



IT - Bordo sensibile

IT - Bordo sensibile

IT - Bordo sensibile





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Per verificare le parti della centrale vedere **fig. 2**.

Legenda fig. 2:

- A** Connettore alimentazione 24 Vac
- B** Connettore per batteria tampone PS124 / sistema di alimentazione a energia solare Solemyo (per approfondimenti vedere capitolo 6.3)
- C** Fusibile servizi (2A, tipo F)
- D** Innesco "SM" per ricevitore radio
- E** Morsetto motore M1 (parte per primo in chiusura)
- F** Morsetto motore M2 (parte per primo in apertura)
- G** Morsetto uscita lampeggiante
- H** Morsetto uscita OGI o elettroserratura
- I** Morsetti 24 Vdc per servizi e fototest
- L9..L13** Led ingressi
- OK** Led di stato "Led Ok"
- L1..L8** Led programmazione
- LR** Led programmazione radio
- M** Morsetti per ingressi
- N** Morsetto per antenna radio
- O** Selettore motore
- Q** Innesco per IBT4N
- R** Fusibile di rete (vedi: caratteristiche tecniche del prodotto)
- S** Alimentazione di rete (L-Linea; N-Neutro) (vedi: caratteristiche tecniche del prodotto)
- T** Connessione di terra
- U** Pressacavo
- P1..P3** Pulsanti per programmazione centrale
- P4** Pulsante per programmazione radio

2.1 - Verifiche preliminari all'installazione

Prima di procedere all'installazione, è necessario verificare l'integrità dei componenti del prodotto, l'adeguatezza del modello scelto e l'idoneità dell'ambiente destinato all'installazione:

- Verificare che tutte le condizioni di utilizzo rientrino nei "limiti d'impiego" e nelle "Caratteristiche tecniche" del prodotto".
- Verificare che l'ambiente scelto per l'installazione sia compatibile con l'ingombro totale del prodotto (**fig. 3**).
- Verificare che la superficie scelta per l'installazione del prodotto sia solida e possa garantire un fissaggio stabile.
- Verificare che la zona di fissaggio non sia soggetta ad allagamenti; prevedere il montaggio del prodotto adeguatamente sollevato da terra.
- Verificare che lo spazio intorno al prodotto consenta una facile e sicura esecuzione delle manovre manuali.
- Verificare che nell'automazione siano presenti gli arresti meccanici sia in Chiusura sia in Apertura.

2.2 - Limiti d'impiego del prodotto

Il prodotto può essere utilizzato esclusivamente con i motoriduttori WG2024, WG3524, WG4024, WG5024, TOO3024, TOO4524, XME2024.

2.3 - Installazione

Per il fissaggio della Centrale, procedere come mostrato in **fig. 4**. Inoltre, rispettare le seguenti avvertenze:

- La centrale viene fornita in un contenitore che, se adeguatamente installato, garantisce un grado di protezione classificato IP54. Pertanto la centrale è adatta ad essere installata anche all'esterno.
- Fissare la centrale su una superficie irremovibile, verticale, piana e adeguatamente protetta da possibili urti. **Attenzione!** – La parte inferiore della centrale deve distare almeno 40 cm dal terreno.
- Inserire gli appositi passacavi o passatubi nella parte inferiore del contenitore (fig. 4). **Attenzione!** – Se i tubi di protezione dei cavi terminano in un pozzetto, è probabile che all'interno del contenitore della centrale si formi della condensa, danneggiando così la scheda elettronica. In questo caso, proteggere adeguatamente la centrale in modo da prevenire la formazione di condensa.
- È possibile inserire i passacavi sul lato lungo del contenitore soltanto se la centrale viene installata all'interno, in ambiente protetto.

Per eseguire l'installazione degli altri dispositivi presenti nell'automazione, fare riferimento ai rispettivi manuali d'istruzioni.

2.4 - Collegamenti elettrici

ATTENZIONE!

– **Tutti i collegamenti elettrici devono essere eseguiti in assenza di alimentazione elettrica di rete e con la batteria tampone scollegata, se presente nell'automazione.**

– **Le operazioni di collegamento devono essere eseguite esclusivamente da personale qualificato.**

– **Verificare che tutti i cavi elettrici da utilizzare siano del tipo adatto.**

- 01.** Svitare le viti del coperchio;
 - 02.** Predisporre i fori per il passaggio dei cavi elettrici;
 - 03.** Eseguire i collegamenti dei cavi facendo riferimento allo schema elettrico di **fig. 5**. Per collegare il cavo dell'alimentazione elettrica, vedere **fig. 2** e **fig. 6** nei punti R, S, T, U. Per il bloccaggio del cavo utilizzare il **pressacavo U**.
- Nota** – Per facilitare i collegamenti dei cavi, è possibile estrarre i morsetti dalle proprie sedi.
- Gli ingressi dei contatti di tipo NC (Normalmente Chiuso), se non usati, vanno ponticellati con "COMUNE" (escluso gli ingressi delle fotocellule se viene attivata la funzione FOTOTEST, per chiarimenti vedere il paragrafo 2.4.3).
 - Se per lo stesso ingresso ci sono più contatti NC vanno posti in SERIE tra di loro.
 - Gli ingressi dei contatti di tipo NA (Normalmente Aperto) se non usati vanno lasciati liberi.
 - Se per lo stesso ingresso ci sono più contatti NA vanno posti in PARALLELO tra di loro.
 - I contatti devono essere assolutamente di tipo elettromeccanico e svincolati da qualsiasi potenziale, non sono ammessi collegamenti a stadi tipo quelli definiti "PNP", "NPN", "Open Collector" ecc.
 - Nel caso di ante sovrapposte è possibile selezionare quale motore deve partire per primo in apertura collegandolo al morsetto M2.

Legenda delle figs. 2 - 5a - 5b - 5c:

Morsetti	Funzione	Descrizione	Tipo cavo
L - N -	Linea di alim.	Alimentazione da rete	3 x 1,5 mm ²
M	Motore 1	Collegamento del motore M1 (Nota 1)	3 x 1,5 mm ²
M	Motore 2	Collegamento del motore M2	3 x 1,5 mm ²
1÷2	Lampeggiante	Collegamento del lampeggiante 24 V $\overline{\text{---}}$ max 25 W	2 x 1 mm ²
3÷4	OGI / Elettroserratura	Collegamento per Spia Cancellato Aperto 24 V $\overline{\text{---}}$ max 5 W o Elettroserratura 12 V $\overline{\text{---}}$ max 25 VA (vedere capitolo 5 - Programmazione)	OGI: 2 x 0,5 mm ² Elettroserratura: 2 x 1 mm ²
5	Comune 24 V $\overline{\text{---}}$ (con Stand by tutto / fototest)	Alimentazione +24 V $\overline{\text{---}}$ per TX fotocellule con fototest (max 100 mA); "COMUNE" per tutti gli ingressi di sicurezza, con funzione "Stand by tutto" attiva (Nota 2)	1 x 0,5 mm ²
6	0 V $\overline{\text{---}}$	Alimentazione 0 V $\overline{\text{---}}$ per servizi	1 x 0,5 mm ²
7	24 V $\overline{\text{---}}$	Alimentazione servizi, senza "Stand by tutto" (24 V $\overline{\text{---}}$ max 200 mA)	1 x 0,5 mm ²
8	Comune 24 V $\overline{\text{---}}$	Comune per tutti gli ingressi (+24 V $\overline{\text{---}}$) senza "Stand by tutto"	1 x 0,5 mm ²
9	ALT	Ingresso con funzione di ALT (emergenza, blocco di sicurezza) (Nota 3)	1 x 0,5 mm ²
10	FOTO	Ingresso NC per dispositivi di sicurezza (fotocellule, bordi sensibili)	1 x 0,5 mm ²
11	FOTO1	Ingresso NC per dispositivi di sicurezza (fotocellule, bordi sensibili)	1 x 0,5 mm ²
12	PASSO-PASSO	Ingresso per funzionamento ciclico (APRE-STOP-CHIUDE-STOP)	1 x 0,5 mm ²
13	AUX	Ingresso ausiliario (Nota 4)	1 x 0,5 mm ²
	Antenna	Collegamento antenna del ricevitore radio	cavo schermato tipo RG58

- Nota 1** – Non usato per cancelli ad una sola anta (la centrale riconosce automaticamente se c'è un solo motore installato)
- Nota 2** – La funzione "Stand by tutto" serve per ridurre i consumi; per approfondimenti sui collegamenti elettrici vedere paragrafo 2.4.1 "Collegamento Stand by tutto/Fototest" e per la programmazione vedere capitolo 5.2 "Funzioni programmabili".
- Nota 3** – L'ingresso ALT può essere utilizzato per contatti NC oppure a resistenza costante 8,2 K Ω in auto apprendimento (vedere capitolo "Programmazione")
- Nota 4** – L'ingresso ausiliario AUX di fabbrica è programmato con la funzione "Apri parziale tipo 1", ma può essere programmato con una delle seguenti funzioni:

Funzione	Tipo ingresso	Descrizione
APRE PARZIALE TIPO 1	NA	Apri completamente l'anta superiore
APRE PARZIALE TIPO 2	NA	Apri le 2 ante fino a metà della corsa